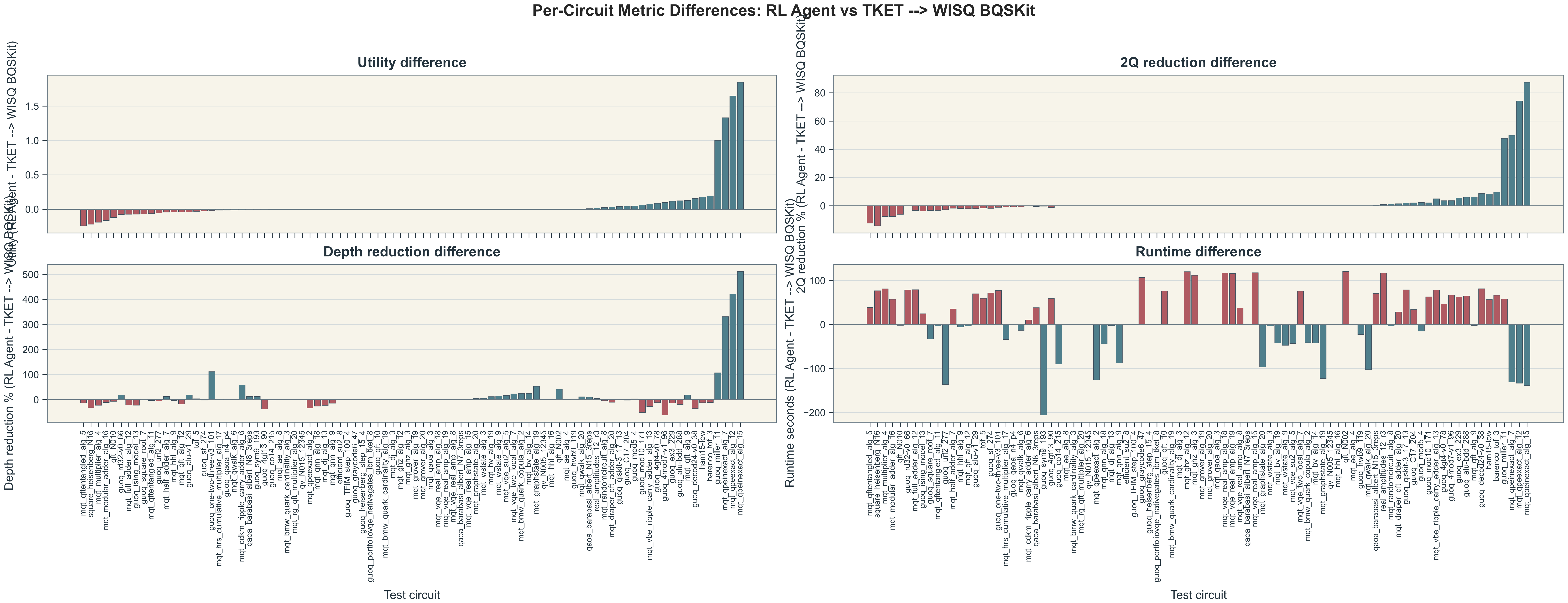
Per-Circuit Metric Differences: RL Agent vs TKET --> WISQ BQSKit
Utility difference
Utility (RL Agent - TKET --> WISQ BQSKit)
1.5
1.0
0.5
0.0
2Q reduction difference
2Q reduction % (RL Agent - TKET --> WISQ BQSKit)
80
60
40
20
0
Depth reduction difference
Depth reduction % (RL Agent - TKET --> WISQ BQSKit)
500
400
300
200
100
0
Runtime difference
Runtime seconds (RL Agent - TKET --> WISQ BQSKit)
100
0
−100
−200
Test circuit
mqt_qftentangled_alg_5
square_heisenberg_N16
mqt_multiplier_alg_4
mqt_modular_adder_alg_16
qft_N010
guoq_rd32-v0_66
mqt_full_adder_alg_12
guoq_ising_model_13
guoq_square_root_7
mqt_qftentangled_alg_11
guoq_urf2_277
mqt_half_adder_alg_7
mqt_hhl_alg_9
mqt_qft_alg_12
guoq_alu-v1_29
tof_5
guoq_sf_274
guoq_one-two-three-v3_101
mqt_hrs_cumulative_multiplier_alg_17
guoq_qaoa_n4_p4
mqt_qwalk_alg_6
mqt_cdkm_ripple_carry_adder_alg_6
qaoa_barabasi_albert_N8_3reps
guoq_sym9_193
guoq_4gt13_90
guoq_co14_215
mqt_ae_alg_8
mqt_bmw_quark_cardinality_alg_3
mqt_rg_qft_multiplier_alg_20
qv_N015_12345
mqt_qpeexact_alg_2
mqt_qnn_alg_18
mqt_dj_alg_13
mqt_qnn_alg_9
efficient_su2_8
guoq_TFIM_step_100_4
guoq_graycode6_47
guoq_heisenberg_step_15_4
guoq_portfoliovqe_nativegates_ibm_tket_8
guoq_qft_10
mqt_bmw_quark_cardinality_alg_19
mqt_dj_alg_3
mqt_ghz_alg_12
mqt_ghz_alg_3
mqt_grover_alg_19
mqt_grover_alg_20
mqt_qaoa_alg_3
mqt_vqe_real_amp_alg_18
mqt_vqe_real_amp_alg_19
mqt_vqe_real_amp_alg_8
qaoa_barabasi_albert_N7_3reps
mqt_vqe_real_amp_alg_15
mqt_graphstate_alg_20
mqt_wstate_alg_3
mqt_bv_alg_19
mqt_wstate_alg_9
mqt_vqe_su2_alg_5
mqt_vqe_two_local_alg_7
mqt_bmw_quark_copula_alg_2
mqt_bv_alg_14
mqt_graphstate_alg_19
qv_N005_12345
mqt_hhl_alg_16
qft_N002
mqt_ae_alg_4
guoq_hwb9_119
mqt_qwalk_alg_20
qaoa_barabasi_albert_N15_3reps
real_amplitudes_12_r3
mqt_randomcircuit_alg_8
mqt_draper_qft_adder_alg_20
guoq_qiskit-3_17_13
guoq_C17_204
guoq_mod5_4
guoq_mod10_171
mqt_vbe_ripple_carry_adder_alg_13
guoq_4gt4-v0_78
guoq_4mod7-v1_96
guoq_ex3_229
guoq_alu-bdd_288
mqt_qft_alg_9
guoq_decod24-v0_38
ham15-low
barenco_tof_3
guoq_miller_11
mqt_qpeinexact_alg_7
mqt_qpeexact_alg_12
mqt_qpeinexact_alg_15
Test circuit
mqt_qftentangled_alg_5
square_heisenberg_N16
mqt_multiplier_alg_4
mqt_modular_adder_alg_16
qft_N010
guoq_rd32-v0_66
mqt_full_adder_alg_12
guoq_ising_model_13
guoq_square_root_7
mqt_qftentangled_alg_11
guoq_urf2_277
mqt_half_adder_alg_7
mqt_hhl_alg_9
mqt_qft_alg_12
guoq_alu-v1_29
tof_5
guoq_sf_274
guoq_one-two-three-v3_101
mqt_hrs_cumulative_multiplier_alg_17
guoq_qaoa_n4_p4
mqt_qwalk_alg_6
mqt_cdkm_ripple_carry_adder_alg_6
qaoa_barabasi_albert_N8_3reps
guoq_sym9_193
guoq_4gt13_90
guoq_co14_215
mqt_ae_alg_8
mqt_bmw_quark_cardinality_alg_3
mqt_rg_qft_multiplier_alg_20
qv_N015_12345
mqt_qpeexact_alg_2
mqt_qnn_alg_18
mqt_dj_alg_13
mqt_qnn_alg_9
efficient_su2_8
guoq_TFIM_step_100_4
guoq_graycode6_47
guoq_heisenberg_step_15_4
guoq_portfoliovqe_nativegates_ibm_tket_8
guoq_qft_10
mqt_bmw_quark_cardinality_alg_19
mqt_dj_alg_3
mqt_ghz_alg_12
mqt_ghz_alg_3
mqt_grover_alg_19
mqt_grover_alg_20
mqt_qaoa_alg_3
mqt_vqe_real_amp_alg_18
mqt_vqe_real_amp_alg_19
mqt_vqe_real_amp_alg_8
qaoa_barabasi_albert_N7_3reps
mqt_vqe_real_amp_alg_15
mqt_graphstate_alg_20
mqt_wstate_alg_3
mqt_bv_alg_19
mqt_wstate_alg_9
mqt_vqe_su2_alg_5
mqt_vqe_two_local_alg_7
mqt_bmw_quark_copula_alg_2
mqt_bv_alg_14
mqt_graphstate_alg_19
qv_N005_12345
mqt_hhl_alg_16
qft_N002
mqt_ae_alg_4
guoq_hwb9_119
mqt_qwalk_alg_20
qaoa_barabasi_albert_N15_3reps
real_amplitudes_12_r3
mqt_randomcircuit_alg_8
mqt_draper_qft_adder_alg_20
guoq_qiskit-3_17_13
guoq_C17_204
guoq_mod5_4
guoq_mod10_171
mqt_vbe_ripple_carry_adder_alg_13
guoq_4gt4-v0_78
guoq_4mod7-v1_96
guoq_ex3_229
guoq_alu-bdd_288
mqt_qft_alg_9
guoq_decod24-v0_38
ham15-low
barenco_tof_3
guoq_miller_11
mqt_qpeinexact_alg_7
mqt_qpeexact_alg_12
mqt_qpeinexact_alg_15